Machine Learning-Enhanced EMA Trading Strategy

European Equity Markets (2021–2025)

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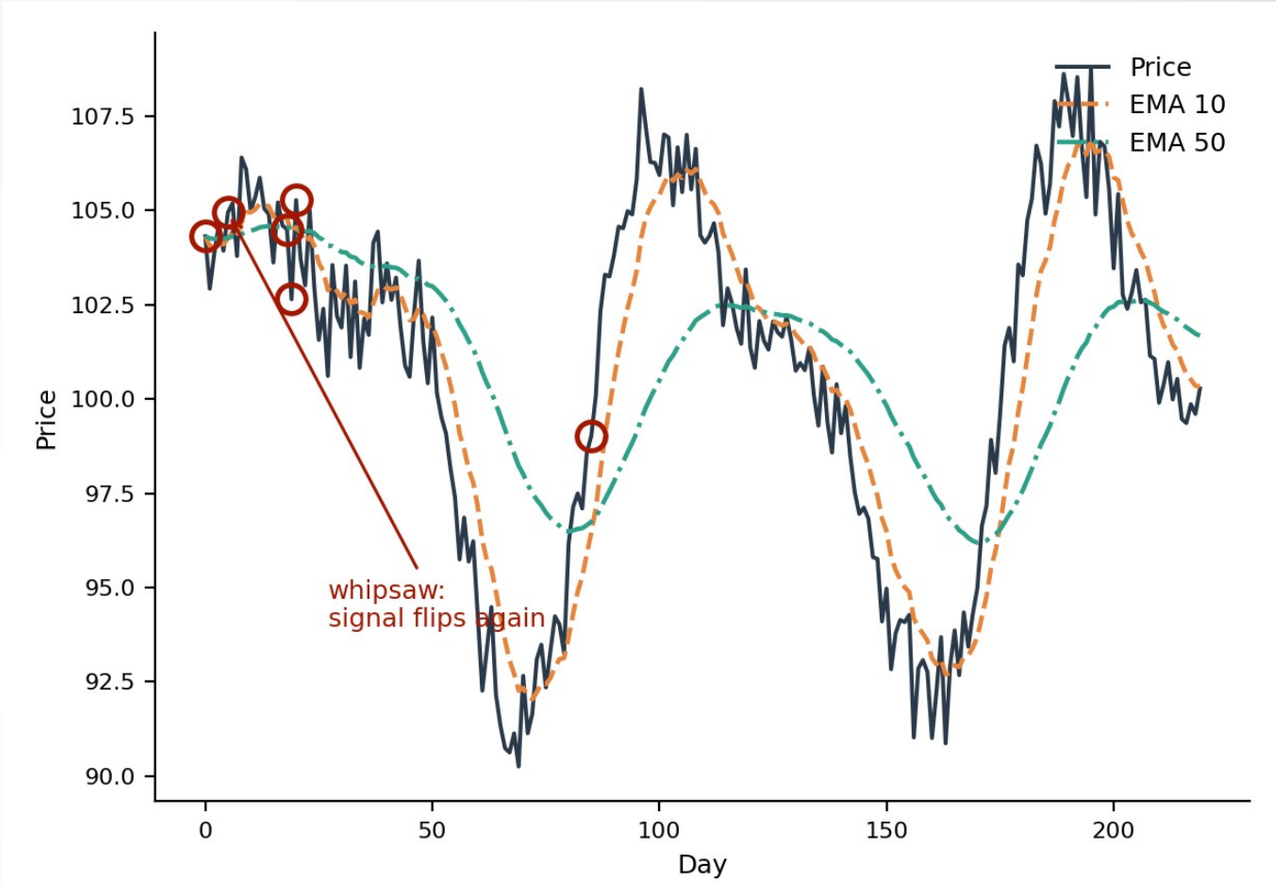
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2026 DEDA Digital Economy Data Analytics, Group Work

Why Does a Classic EMA Strategy Need Help?

- EMA crossovers offer a transparent, easy-to-implement way to follow price trends.
- But in sideways markets, frequent trend reversals create false signals (whipsaws) that hurt performance.
- Idea: add a machine-learning filter that skips trades when the estimated direction is uncertain.
- Goal: improve risk-adjusted outcomes — not prediction accuracy alone.



Schematic: in a range market the two EMAs cross repeatedly, and each flip trades the wrong side first.

Research Questions and Contributions

- RQ1 Does an RF confidence filter outperform the traditional EMA strategy in European equities?
- RQ2 Which technical indicators matter most for next-day return prediction?
- RQ3 Are the results stable across filtering rules, transaction costs, and portfolio-neutrality definitions?

Contribution A chronological (walk-forward) evaluation of the RF-filtered EMA strategy, with transparent diagnostics and robustness checks.

Outlines

- **1. Motivation and Research Questions**
- **2. Data and Descriptive Statistics**
  - 2.1 Ten European Country ETFs
  - 2.2 Market Landscape and Return Correlations
- **3. Methodology**
  - 3.1 Technical Features: 25 Candidates
  - 3.2 EMA Direction and Random Forest Prediction
  - 3.3 Confidence Filtering and Portfolio Construction
- **4. Out-of-Sample Results and Robustness**
- **5. Conclusions and Limitations**



Dataset: Ten European Country ETFs

- Ten US-listed iShares MSCI country ETFs, quoted in USD.
- Daily bars from 2021-01-04 to 2025-12-31, one row per ETF-day.

Date	Ticker	Market	Open	High	Low	Close	Adj Close	Volume
2021-01-04	EWD	Sweden	40.97	40.97	40.03	40.30	33.48	206,700
2021-01-04	EWG	Germany	32.64	32.66	31.93	32.16	27.88	4,783,800
2021-01-04	EWQ	France	34.20	34.25	33.46	33.63	28.10	786,300

Source file: European_ETF_Daily_2021_2025.csv

EWD Sweden	**EWN** Netherlands
EWG Germany	**EWO** Austria
EWI Italy	**EWP** Spain
EWK Belgium	**EWQ** France
EWL Switzerland	**EWU** United Kingdom

Data Quality and Chronological Split

- **Data quality:** 12,550 raw ETF-day rows; no missing values in required fields; no duplicated Date × Ticker pairs.
- **Chronological split:** a time series is never randomly shuffled; training labels must be realized before the next period begins (boundary purge).

Training
2021–2023
7,320 ETF-days

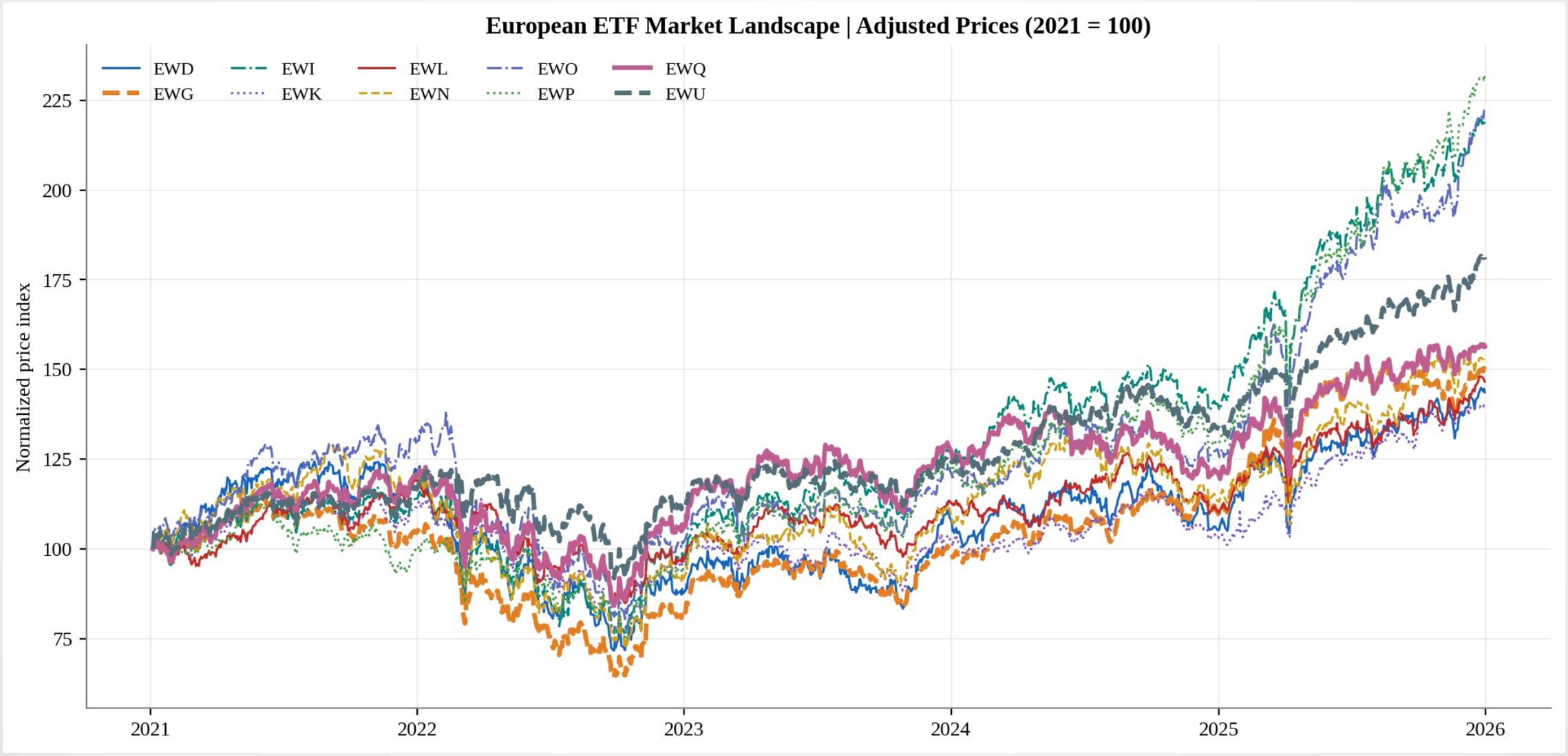
Validation
2024
2,510 ETF-days

Test
2025
2,490 rows · 249 dates

Split counts are model-ready ETF-days: after indicator warm-up, 12,320 of the 12,550 rows carry complete features.

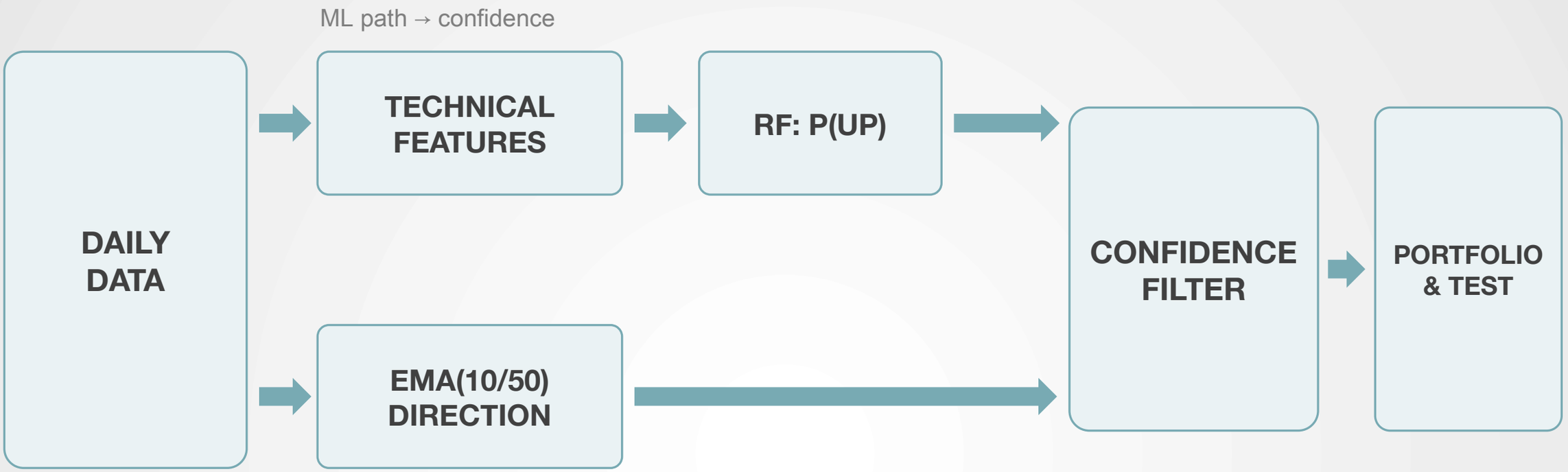
European Equity Market Landscape

- **First, the big picture:** broad swings are shared across the ten markets — a deep drawdown around 2022, then a strong recovery in 2023–2025.



KEY READING Adjusted ETF prices, normalized to 100 at the first observation.

End-to-End Research Workflow



Two parallel paths: EMA(10/50) supplies direction; Random Forest supplies confidence.

2021–2023
Training

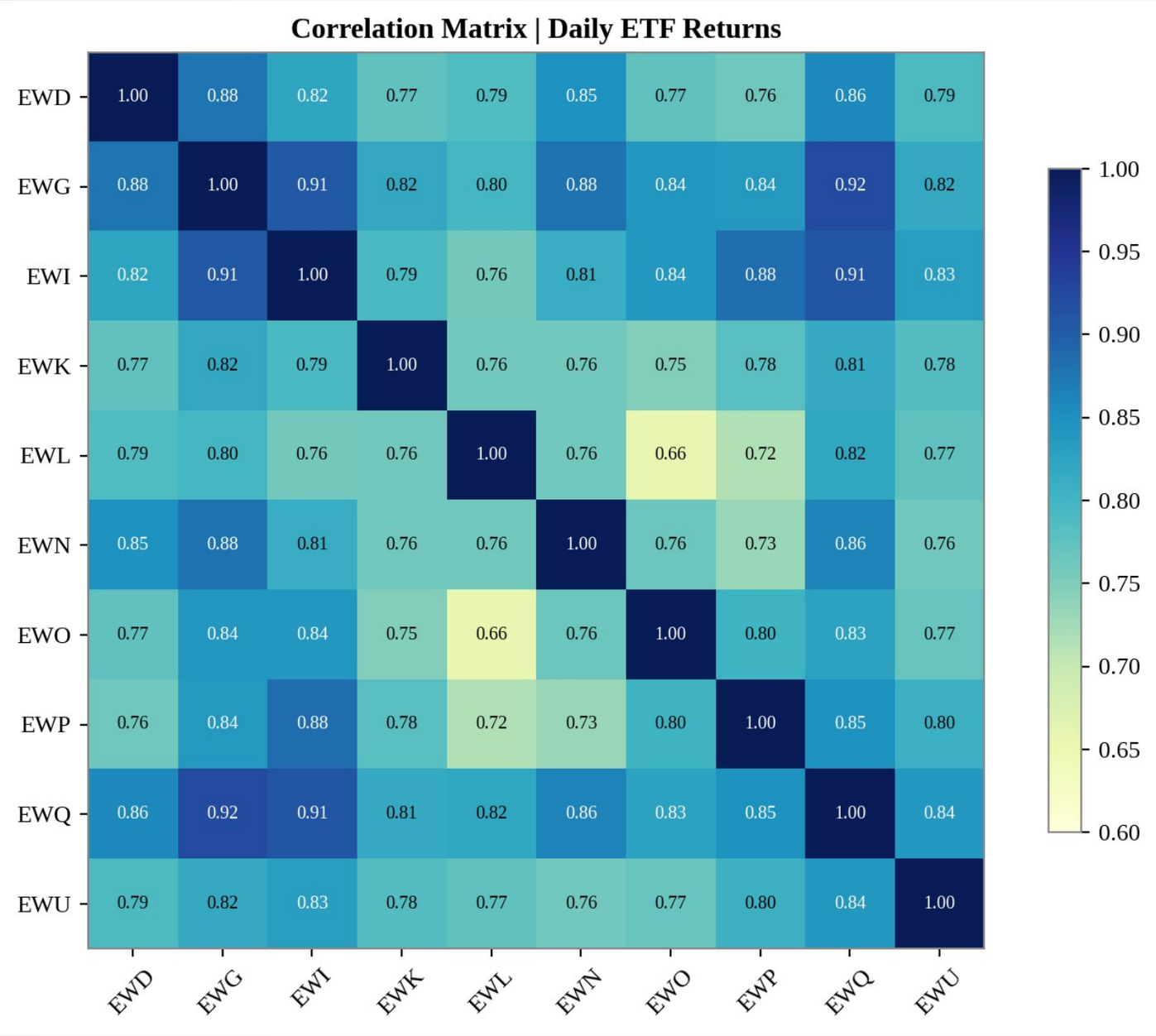
2024
Threshold Validation

2025
Out-of-Sample Test

Technical Features: 25 Candidates

- **Momentum (4):** returns over the last 1, 5, 10, and 20 days
  - ret_1d, ret_5d, momentum_10d, momentum_20d
- **Mean reversion (4):** RSI(14), price Z-score(20), Bollinger position and width
  - rsi_14, zscore_20, bb_position_20, bb_width_20
- **Trend (4):** fast/slow EMA levels, EMA gap, and gap to the 20-day mean
  - ema_10, ema_50, ema_gap_10_50, price_ma20_gap
- **MACD (3):** MACD line, signal line, and histogram, each scaled by price
  - macd_pct, macd_signal_pct, macd_hist_pct
- **Volatility & risk (6):** 10/20-day volatility, volatility ratio, ATR(14), skewness, drawdown
  - vol_10d, vol_20d, vol_ratio_10_20, atr_14_pct, skew_20, drawdown_20
- **Volume & intraday info (4):** volume Z-score and ratio, overnight gap, intraday return
  - volume_z_20, volume_ratio_5_20, overnight_gap, intraday_return
- All indicators are computed within each ETF features at date t only use information up to t (no look-ahead).

Return Correlations: Diversification Is Limited



Pairwise daily-return correlations range roughly from **0.66 to 0.92**: the country ETFs share common European factors.

Diversification across the ETFs is therefore limited, and the strategy relies on timing within each market.

KEY READING Lowest pair: EWL–EWO (Switzerland–Austria) at 0.66; highest: EWG–EWQ (Germany–France) at 0.92.

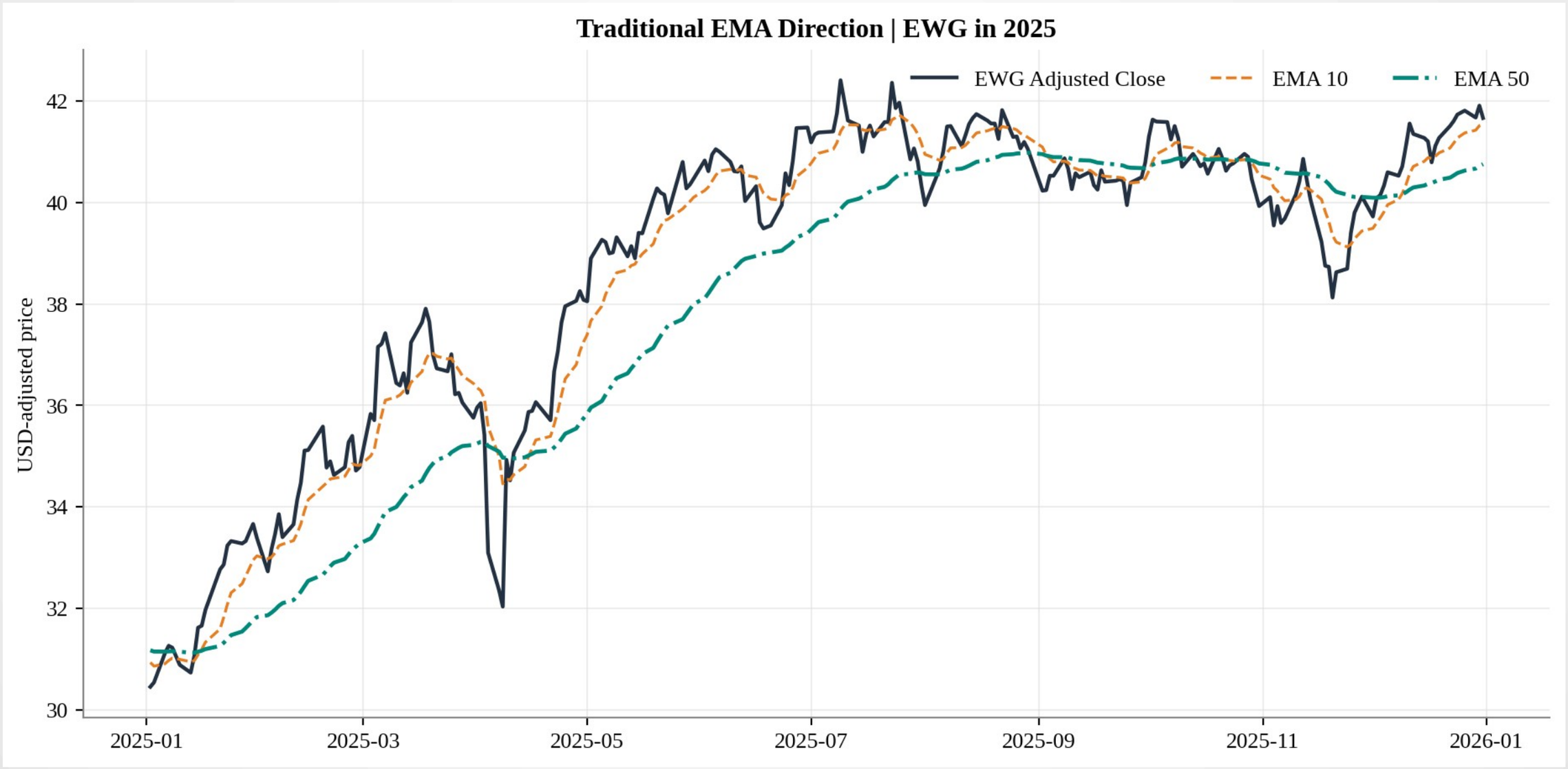
Code Available on GitHub

- Everything needed to reproduce this study lives in one repository: data preparation, feature engineering, model training, and the backtest.
- **Repository contents:**
  - European_ETF_Daily_2021_2025.csv — daily OHLCV data for the ten ETFs
  - European_ML_EMA_Concise.py — features, Random Forest, signals, backtest
  - Generate_Research_Figures.py — all figures used in this deck
  - requirements.txt — Python environment
- **Quick start:** pip install -r requirements.txt, then python3 European_ML_EMA_Concise.py



EMA Determines Long or Short Direction

EMA(10) ≥ EMA(50) → direction = +1 (long); otherwise → –1 (short).



KEY READING German ETF EWG in 2025. Crossovers mark regime changes; rapid flips in flat stretches are where whipsaw losses come from.

Random Forest: Feature Selection and Prediction

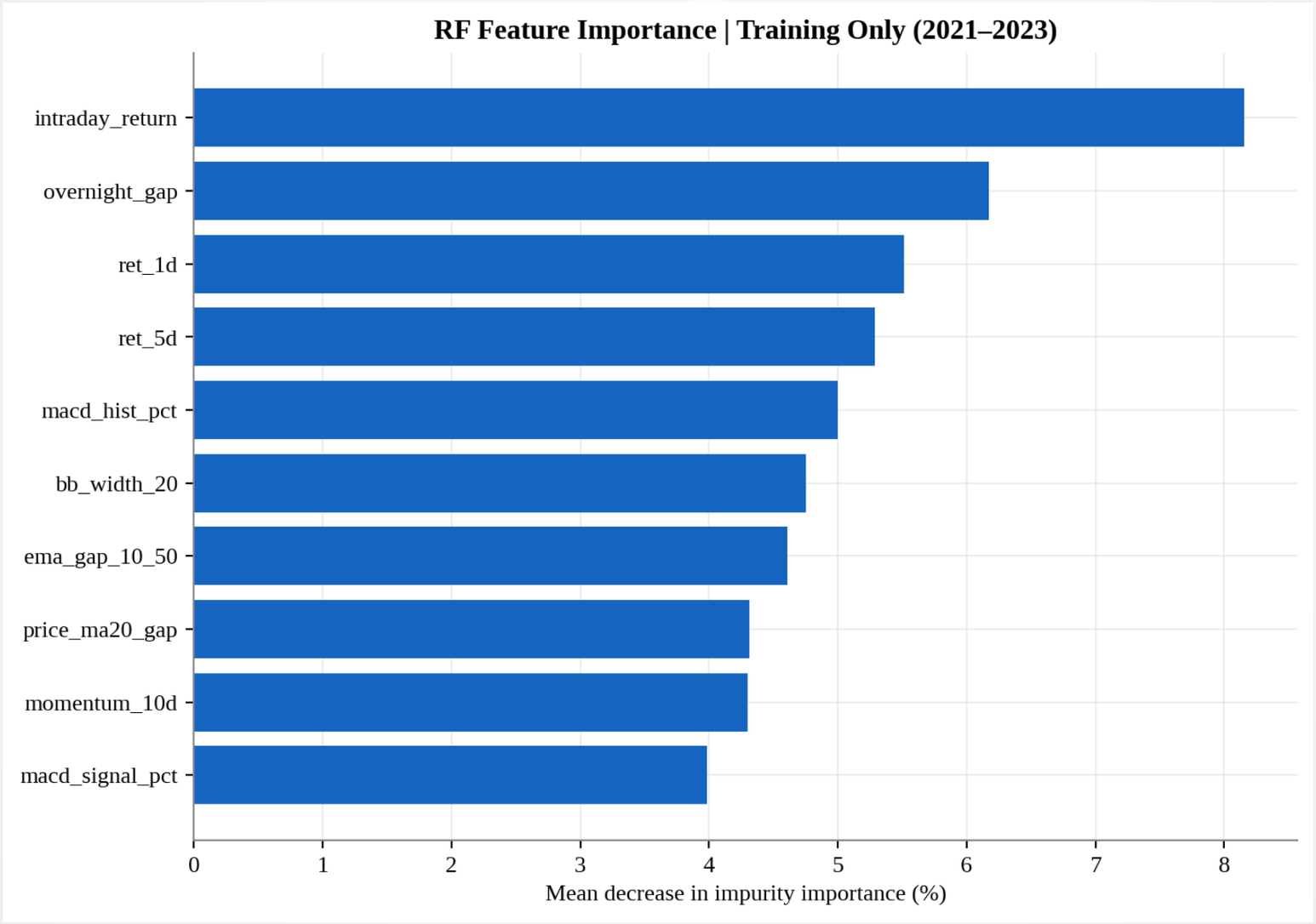
- **Binary target:** y = 1 if the next-day adjusted-close return is positive, else 0.
- **Model:** Random Forest with 300 trees, maximum depth 6, minimum leaf size 12.
- **Selection:** rank the 25 features by RF importance on 2021–2023 only; keep the top 5.
- **Walk-forward:** fit on 2021–2023 and validate the threshold in 2024; refit through 2024 to predict 2025.

Top 5 features:

- Intraday return
- Overnight gap
- 1-day return
- 5-day return
- MACD histogram

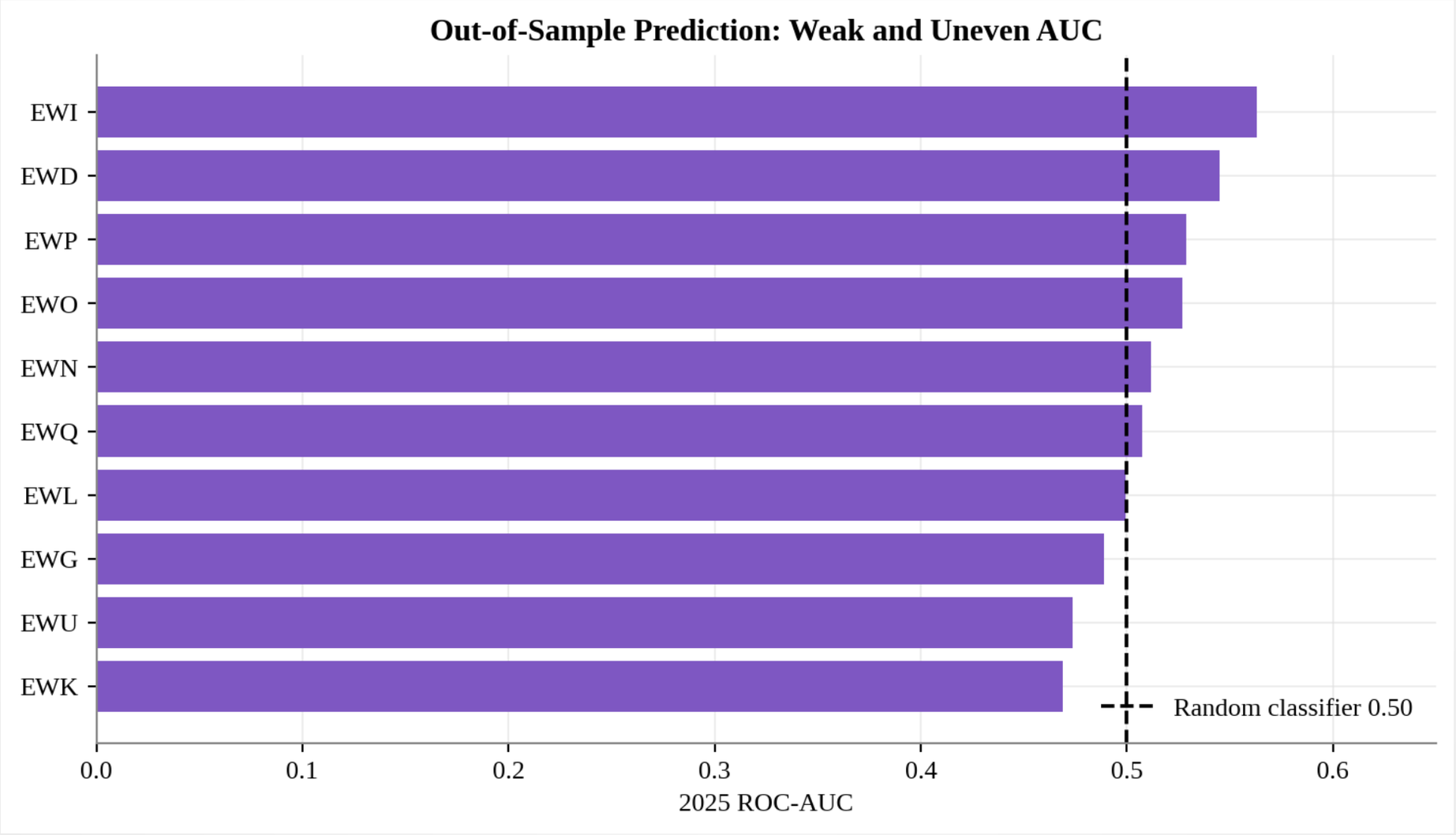
Feature Importance

- Random Forest trained on 2021–2023 data with 25 technical features.
  - Top 5: intraday return, overnight gap, 1-day and 5-day returns, MACD.



Out-of-Sample Prediction

- 2025 test AUC varies across the 10 European-country ETFs.
  - Range: 0.469–0.563; mean: 0.511 (close to 0.50).



Direction-Aware Confidence

- EMA(10/50) determines the trading direction.
  - EMA(10) ≥ EMA(50): long; otherwise: short.

- Random Forest provides the probability of an upward return.
  - Long confidence = P(up); short confidence = 1 − P(up).

- Keep signals only when direction-aware confidence passes the cutoff.
  - Retained positions are scaled by confidence and distance above the cutoff.

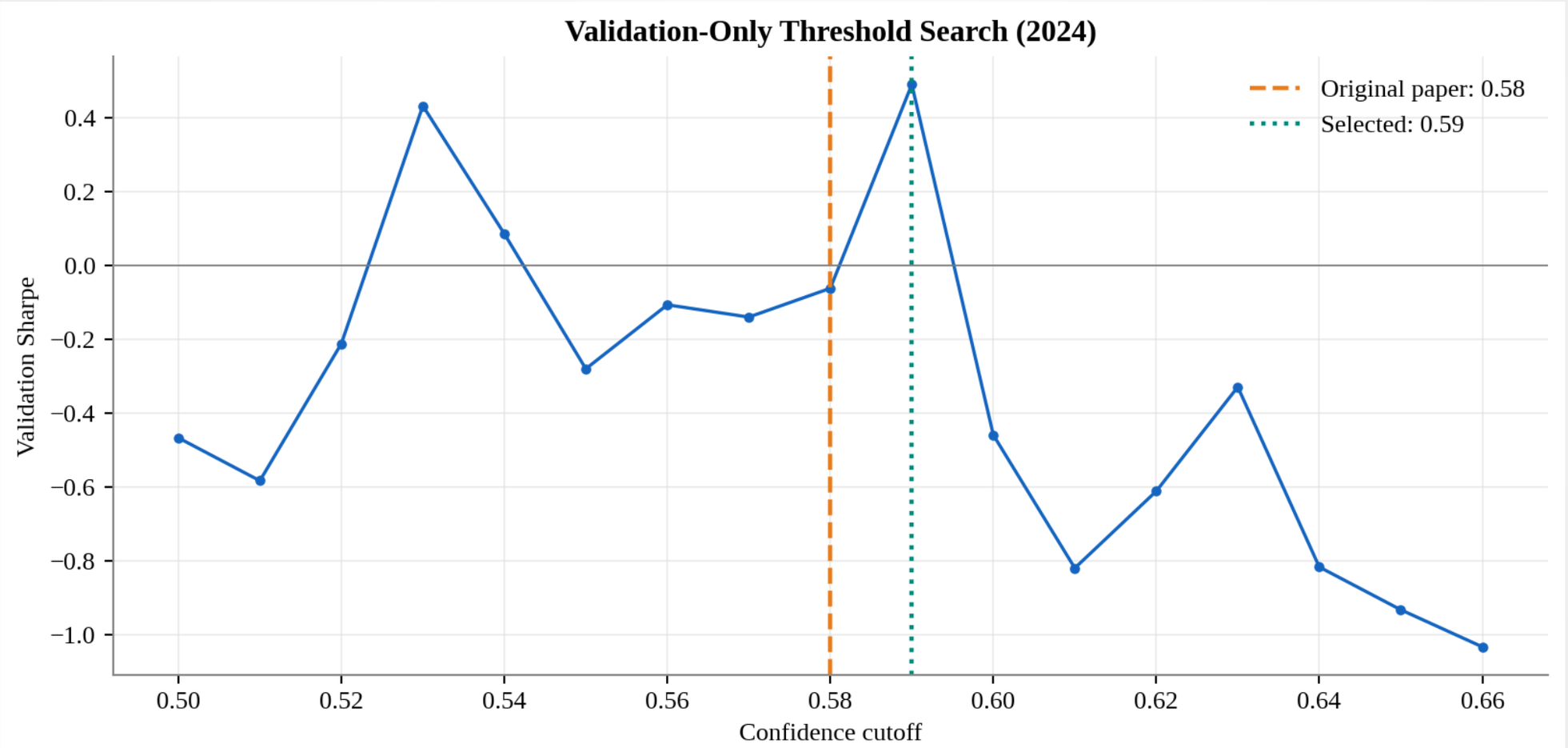
signal = EMA direction × confidence × scaled strength

Three Confidence Filters

- Method 1: Fixed confidence threshold
  - Keep direction-aware signals when confidence ≥ 0.58.
- Method 2: Validation-selected threshold
  - Choose τ = 0.59 using 2024 validation Sharpe; test on 2025.
- Method 3: Dynamic confidence threshold
  - For each ETF: rolling 60-day 90th percentile of earlier confidence.
  - Past observations only; a shifted window prevents current-day look-ahead.

Threshold Validation

- Search τ = 0.50–0.70 using 2024 data, not the 2025 test set.
  - Best validation Sharpe at τ = 0.59: 0.491; performance is uneven.



Portfolio Construction

- Convert selected signals into gross-normalized weights.

$$w(i,t) = s(i,t) / \Sigma_j |s(j,t)| \quad ; \quad \Sigma_i |w(i,t)| = 1 \text{ when active}$$

- Gross-normalized does NOT mean dollar neutral.
  - The sum of portfolio weights need not equal zero.

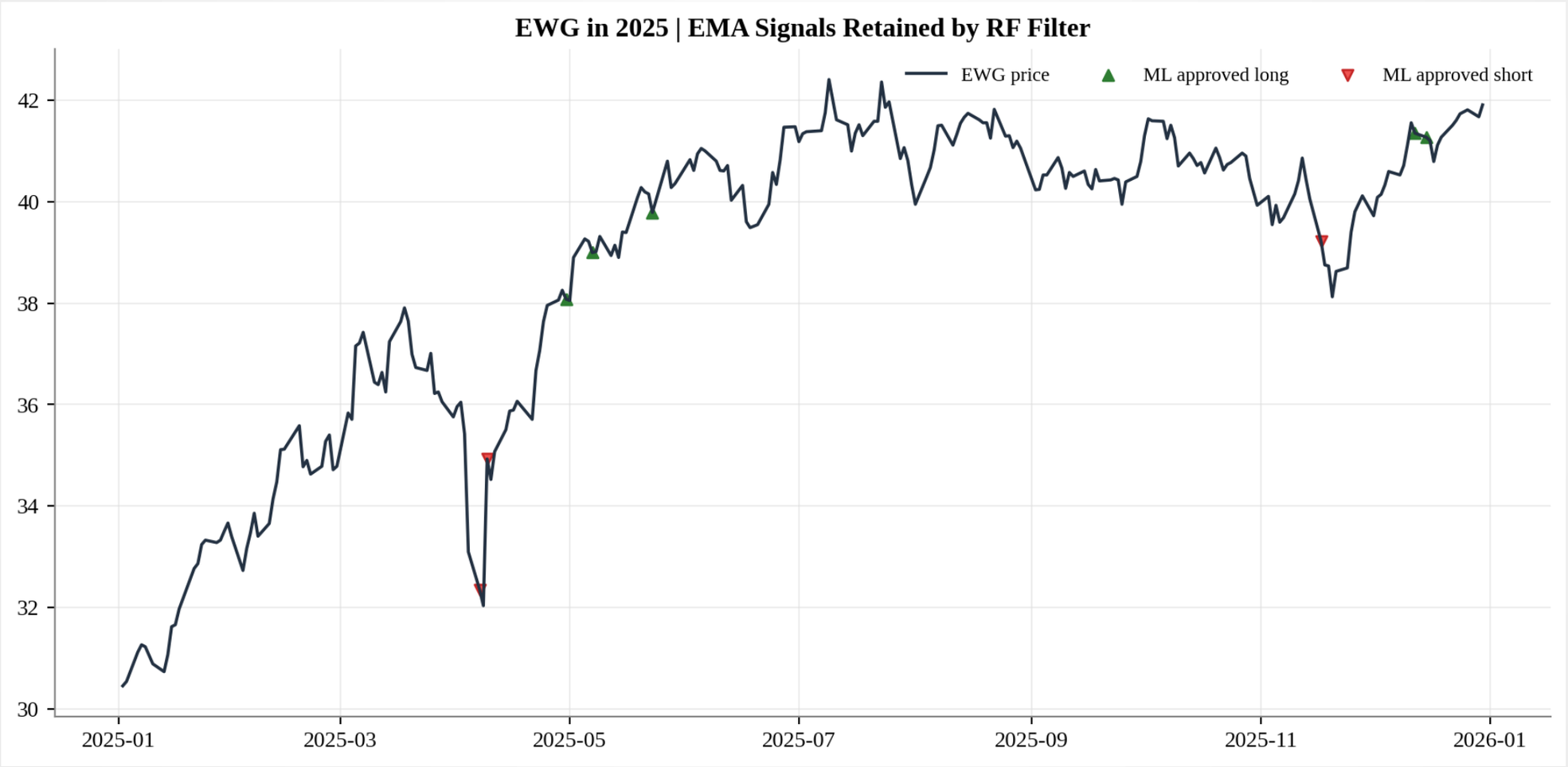
- Compute transaction-cost-adjusted next-day portfolio returns.

$$\text{Turnover}(t) = \tfrac{1}{2} \Sigma_i |w(i,t) - w(i,t-1)|$$

$$\text{Net return} = \text{gross return} - 0.0005 \times \text{turnover}$$

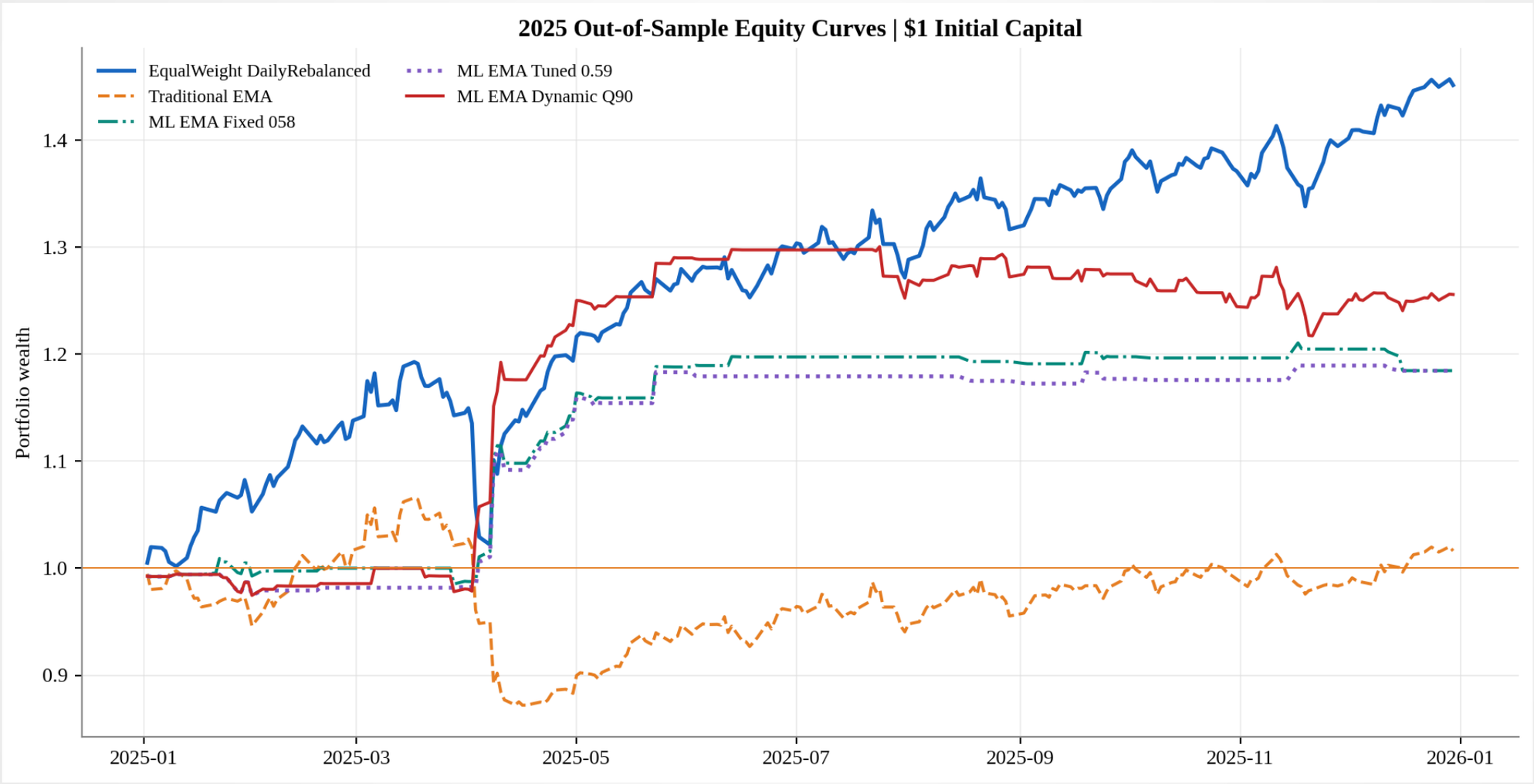
Filtered Trading Signals: EWG

- Example: German equity ETF (EWG) during the 2025 test year.
  - Markers show EMA directions retained after RF confidence filtering.



Cumulative Returns in 2025

- Equal-weight ETF: +45.12%; traditional EMA: +1.62%.
  - ML-EMA fixed: +18.46%; tuned: +18.44%; dynamic: +25.56%.



Performance Comparison

- 2025 out-of-sample returns, risk and trading intensity.

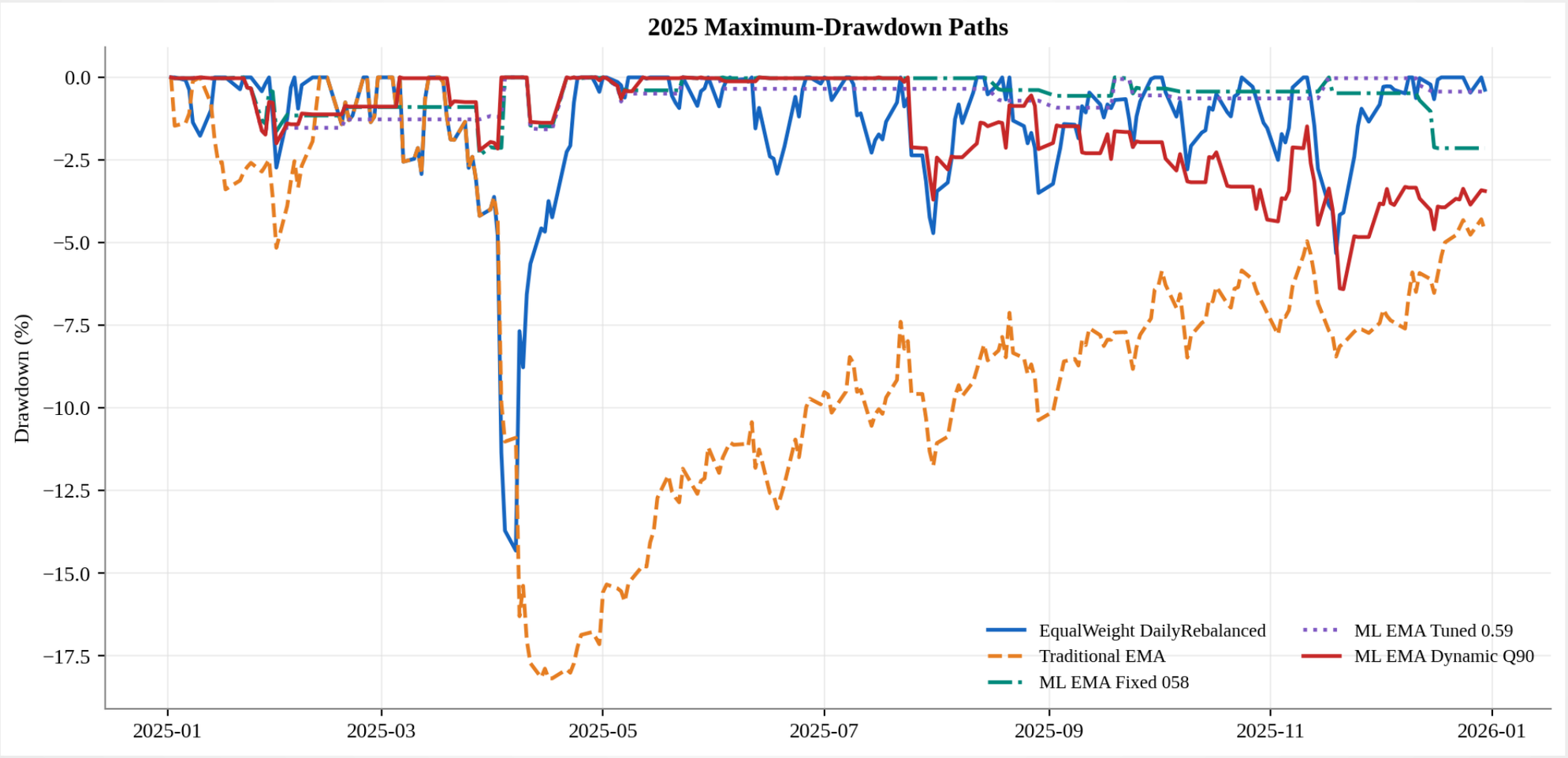
Portfolio	Total return	Volatility	Sharpe	Max DD	Avg. turnover
Equal-weight ETF	45.12%	16.98%	2.31	-14.31%	—
Traditional EMA	1.62%	14.55%	0.19	-18.18%	0.022
ML fixed 0.58	18.46%	10.47%	1.69	-2.36%	0.146
ML tuned 0.59	18.44%	10.25%	1.72	-1.99%	0.125
ML dynamic Q90	25.56%	13.49%	1.77	-6.41%	0.367

- Tuned ML-EMA: Sharpe 1.72, maximum drawdown −1.99%.

Note: Equal-weight returns exclude costs; EMA-based returns include 5 bps per unit of turnover.

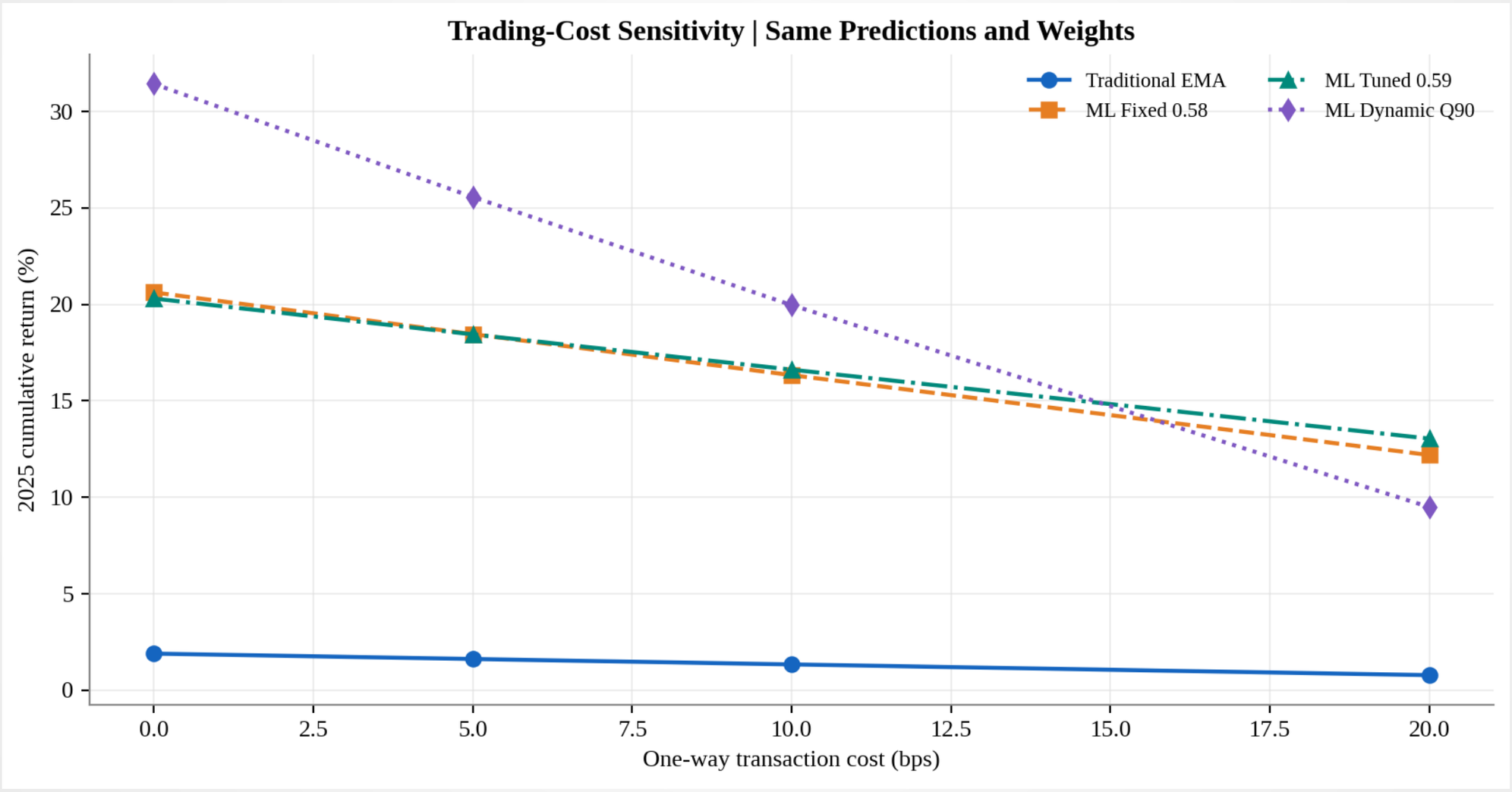
Maximum Drawdown

- Tuned ML-EMA: −1.99%; fixed: −2.36%; traditional EMA: −18.18%.
  - Tuned filter: active on 34/249 days (13.7%); lower drawdown may reflect time in cash.



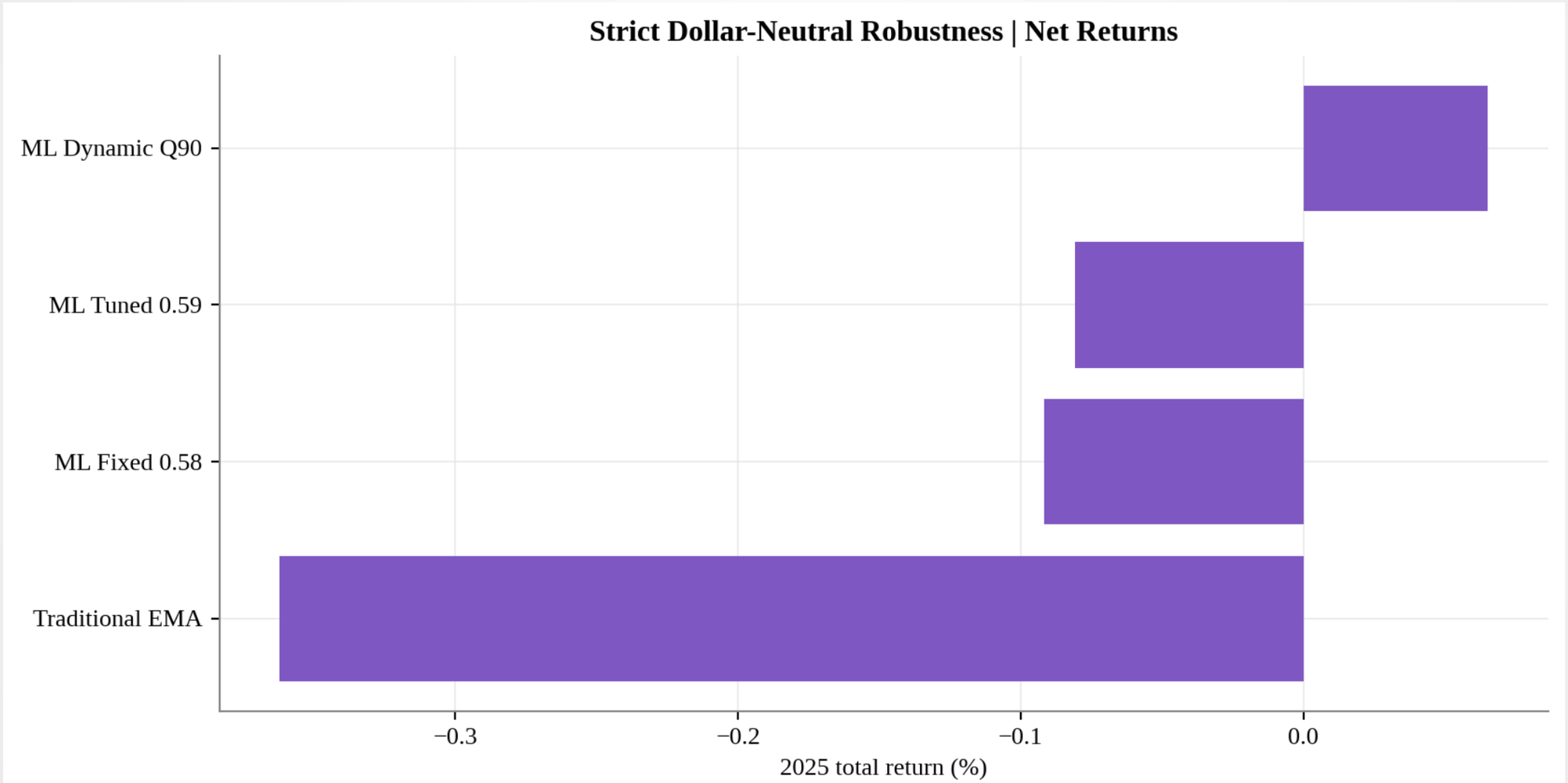
Transaction-Cost Sensitivity

- Compare zero, 5, 10 and 20 basis points per unit of turnover.
  - Dynamic ML-EMA return declines from +31.43% to +9.49%.



Strict Dollar-Neutral Robustness

- Strict neutrality requires +50% long and −50% short exposure.
  - ML rules are active on only 1, 1 and 2 of 249 test dates, respectively.



Conclusions

- 1. The ML filter improves the 2025 results versus traditional EMA.
  - ML returns: +18.44% to +25.56%, versus +1.62% for EMA.
- 2. Lower drawdown does not imply the highest market return.
  - The equal-weight ETF basket leads in total return (+45.12%).
- 3. Predictive ability and market-neutral evidence remain limited.
  - AUC is near 0.50; strict-neutral ML portfolios rarely hold positions.
- 4. Further checks are necessary before practical deployment.
  - Longer walk-forward tests, executable prices, realistic costs and matched benchmarks.

Hyperlink to Github:

https://github.com/uupearl/European-ML-EMA-Trading-Strategy

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